Supplementary Material: Physiology-Guided Temporal Fusion for Cross-Subject EEG–fNIRS Brain–Computer Interfaces

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This supplementary document provides extended experimental analyses, physiological interpretations, and subject-level ablation breakdowns supporting the main manuscript:

1) **Section ??:** Comparative neurovascular coupling dynamics and delay sensitivity analysis between cognitive (mental arithmetic) and sensorimotor (motor imagery) paradigms (Supplementary Fig. **??**).
2) **Section ??:** Subject-by-subject (29 folds) factorial prior retraining breakdown on motor imagery (Supplementary Table **??**).
3) **Section ??:** Mathematical formulation and streaming convergence dynamics of Causal Recursive Euclidean Alignment (CrEA).
4) **Section ??:** Computational reproducibility parameters and training environment specifications (Supplementary Table **??**).
5) **Section ??:** Independent external cohort cross-subject validation on the 15-participant Buccino et al. (2016) benchmark (Supplementary Table **??**).

S.I. Paradigm-Specific Neurovascular Coupling Dynamics

A. Contrast Between Cognitive and Sensorimotor Paradigms

A principal insight revealed by cross-paradigm evaluation (Section V-F of the main manuscript) is that neurovascular coupling manifests distinct functional characteristics across task regimes: metabolic-dominant cognitive processing (Mental Arithmetic: 79.37%; Working Memory: 73.24%) draws decisive synergy from structured hemodynamic delay priors, whereas sensorimotor tasks (Motor Imagery: 66.32%) primarily engage rapid transient electrophysiological desynchronization (μ/β ERD). To systematically characterize this physiological transition and assess delay sensitivity across diverse paradigms, we conducted two dedicated investigations on the 29-participant motor imagery dataset from Shin et al. (2017):

1) **Fixed-Weight Candidate Delay Sweep ($\tau \in [0.0, 8.0]$ s):** Evaluating the sensitivity of the trained cross-modal attention network to varying candidate delay parameters without retraining.
2) **2×2 Factorial Prior Retraining Ablation:** Full model retraining across 29 LOSO folds under four permutations of the Gaussian delay prior $\mathbf{P}$ and the discrete causal mask $\mathbf{M}$ ($P_{\text{on}}, M_{\text{on}}$; $P_{\text{on}}, M_{\text{off}}$; $P_{\text{off}}, M_{\text{on}}$; $P_{\text{off}}, M_{\text{off}}$).

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B. Empirical Findings and Biophysical Mechanism

As illustrated in Supplementary Fig. **??**(a), mental arithmetic exhibits a pronounced delay sensitivity trajectory: accuracy increases monotonically from 75.34% at $\tau = 0.0$ s to plateau across the physiological window ($\tau \in [3.5, 6.5]$ s), reaching 80.46% at $\tau = 8.0$ s ($\Delta = +5.12$ pp, $p < 0.001$). In contrast, motor imagery yields an essentially horizontal trajectory: grand mean accuracy fluctuates between 68.91% ($\tau = 1.0$ s) and 69.60% ($\tau = 8.0$ s), an amplitude range of only 0.69 pp, with overlapping standard error intervals across all candidate latencies.

The retrained factorial decomposition in Supplementary Fig. **??**(b) reinforces this paradigm-dependent disparity. For mental arithmetic, omitting both priors ($P_{\text{off}}, M_{\text{off}}$) collapses accuracy by -4.94 pp (79.37% $\rightarrow$ 74.43%, $p = 6.05 \times 10^{-5}$), with the Gaussian prior accounting for an average marginal contribution of +4.31 pp. Conversely, for motor imagery, retraining with both priors disabled yields 68.28% $\pm$ 13.72% vs. 69.20% $\pm$ 14.40% for the full model ($\Delta = -0.92$ pp, Wilcoxon $W = 106.5$, $p = 0.2128$, non-significant). Omitting causal masking alone ($P_{\text{on}}, M_{\text{off}}$: 68.10%, $p = 0.0978$) or the delay prior alone ($P_{\text{off}}, M_{\text{on}}$: 68.68%, $p = 0.1766$) likewise shows no statistically significant divergence.

Biophysically, this fundamental contrast aligns with established physiological distinctions in functional vascular dynamics reported in neuroimaging literature:

1) **Prefrontal Cognitive Workload (MA):** Sustained 10-second mental arithmetic imposes intense, prolonged metabolic demands on the prefrontal cortex, recruiting deep arteriolar dilation and large venous drainage volume [**?**], [**?**]. This produces robust frontal $[\text{HbO}]$ elevations (typically in the 0.2–0.6 μM range in continuous-wave fNIRS recordings [**?**]) that evolve over 4–7 s and persist throughout task engagement. Consequently, cross-modal attention strongly relies on physiological delay priors to align fast frontal theta oscillations (θ ERS, 4–8 Hz) with delayed capillary hyperoxygenation, preventing acausal over-fitting on held-out participants.
2) **Sensorimotor Lateralization (MI):** Kinesthetic motor imagery elicits immediate contralateral sensorimotor rhythm desynchronization (μ and β ERD, 8–30 Hz) over primary motor cortex (C3/C4). However, imaginary finger

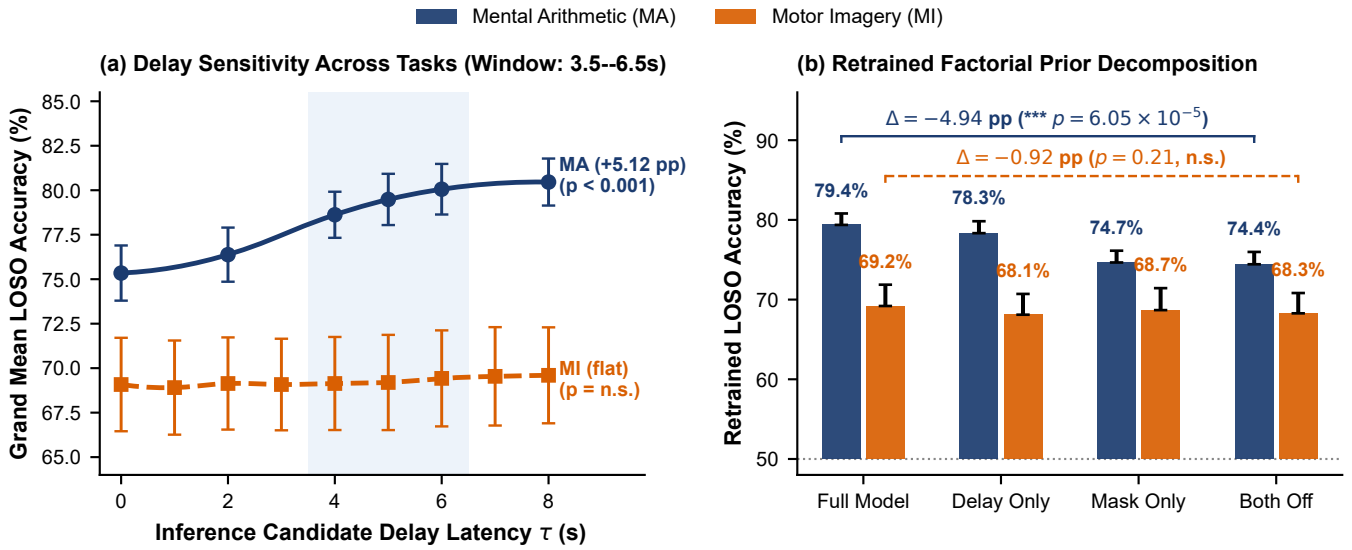


Fig. S1. Comparative neurovascular delay sensitivity and prior ablation contrast between cognitive (Mental Arithmetic, MA) and sensorimotor (Motor Imagery, MI) paradigms across 29 participants under participant-isolated LOSO cross-validation. (a) Candidate delay sensitivity scan across $\tau \in [0.0, 8.0]$ s without retraining. Error bars denote standard error of the mean (SE across $N = 29$ folds). For MA, accuracy rises monotonically from 75.34% at $\tau = 0.0$ s to plateau across the physiological hemodynamic response window ($\tau \in [3.5, 6.5]$ s) and reaches 80.46% at $\tau = 8.0$ s (+5.12 pp, $p < 0.001$), reflecting the sustained 10.0 s task block duration. For MI, the performance curve exhibits essentially delay-invariant behavior within observed experimental variation (ranging between 68.91% and 69.60%, $\Delta \leq 0.69$ pp, $p =$ n.s.). (b) Retrained 2×2 factorial decomposition of delay prior $\mathbf{P}$ and causal mask $\mathbf{M}$. On MA, dropping both priors causes a severe -4.94 pp performance collapse ($p = 6.05 \times 10^{-5}$), confirming the essential regularizing role of neurovascular constraints. On MI, removing both priors decreases accuracy by merely -0.92 pp ($p = 0.2128$, non-significant), indicating that unconstrained attention is sufficient when hemodynamic coupling is weakly localized.

tapping without physical contraction produces significantly attenuated and diffuse hemodynamic changes (typically 0.05–0.15 μM [**?**], [**?**]). Because the localized optical signal-to-noise ratio is lower and neural decodability is dominated by instantaneous spectral ERD, the hemodynamic delay parameter τ imposes minimal regularization constraint, permitting unconstrained cross-attention or direct spatial filtering models (such as FGANet) to perform equally well.

S.II. Subject-by-Subject Factorial Prior Breakdown

To ensure complete empirical transparency, Supplementary Table **??** lists the individual test accuracy achieved across each of the 29 held-out participant folds on motor imagery under all four factorial retraining conditions. Note that this dedicated 29-fold factorial retraining experiment was conducted independently to isolate the individual contributions of the delay prior and causal mask (yielding 69.20% $\pm$ 14.40% for the full model), alongside the initial cross-paradigm benchmark reported in Table V of the main manuscript (66.32% $\pm$ 12.18%). In the dedicated factorial experiment, removing both temporal priors reduced MI accuracy by 0.92 pp ($p = 0.2128$), compared with a 4.94 pp reduction on mental arithmetic.

Between-subject standard deviations ($\sim$ 14.0 pp) significantly exceed the marginal differences observed across model conditions ($\leq$ 1.1 pp). High responders (e.g., Sub-19 and Sub-26: $> 90\%$) decode robustly regardless of prior conditioning, while challenging BCI-illiterate participants (e.g., Sub-10: 45.00%, Sub-24: 50.00%, Sub-12: 51.67%) decode near chance level.

TABLE S1
Participant-by-participant test accuracy (%) across 29 LOSO folds on Motor Imagery under 2×2 factorial retraining of delay prior $\mathbf{P}$ and causal mask $\mathbf{M}$.

Participant Fold	Full Model ($P_{\text{on}}, M_{\text{on}}$)	Mask Only ($P_{\text{off}}, M_{\text{on}}$)	Delay Only ($P_{\text{on}}, M_{\text{off}}$)	Both Off ($P_{\text{off}}, M_{\text{off}}$)	Δ vs. Full (Both Off − Full)
Sub-01	80.00	80.00	81.67	80.00	+0.00
Sub-02	68.33	63.33	65.00	60.00	−8.33
Sub-03	73.33	75.00	73.33	71.67	−1.67
Sub-04	56.67	55.00	60.00	58.33	+1.67
Sub-05	56.67	55.00	56.67	56.67	+0.00
Sub-06	68.33	65.00	68.33	66.67	−1.67
Sub-07	55.00	55.00	53.33	51.67	−3.33
Sub-08	70.00	90.00	73.33	76.67	+6.67
Sub-09	81.67	83.33	81.67	78.33	−3.33
Sub-10	45.00	51.67	46.67	53.33	+8.33
Sub-11	76.67	71.67	73.33	76.67	+0.00
Sub-12	51.67	45.00	45.00	50.00	−1.67
Sub-13	65.00	66.67	63.33	58.33	−6.67
Sub-14	63.33	65.00	65.00	66.67	+3.33
Sub-15	66.67	61.67	65.00	61.67	−5.00
Sub-16	90.00	88.33	85.00	86.67	−3.33
Sub-17	86.67	88.33	83.33	85.00	−1.67
Sub-18	53.33	51.67	53.33	51.67	−1.67
Sub-19	98.33	91.67	91.67	90.00	−8.33
Sub-20	63.33	60.00	63.33	60.00	−3.33
Sub-21	71.67	71.67	70.00	76.67	+5.00
Sub-22	61.67	58.33	61.67	58.33	−3.33
Sub-23	68.33	68.33	68.33	70.00	+1.67
Sub-24	50.00	50.00	48.33	50.00	+0.00
Sub-25	90.00	93.33	93.33	91.67	+1.67
Sub-26	98.33	96.67	98.33	98.33	+0.00
Sub-27	80.00	73.33	71.67	73.33	−6.67
Sub-28	53.33	53.33	51.67	55.00	+1.67
Sub-29	63.33	63.33	63.33	66.67	+3.33
Mean	**69.20**	**68.68**	**68.10**	**68.28**	**−0.92**
SD	14.40	14.85	14.03	13.72	4.10
Macro F1	67.00	66.26	65.79	66.06	−0.94
Wilcoxon p	–	0.1766	0.0978	0.2128	–

Evaluated on the 29-participant Motor Imagery benchmark from Shin et al. (2017) [**?**] across 1,740 trials (60 trials per participant, 30 Left Hand vs. 30 Right Hand). All variants trained from scratch under identical hyperparameter schedules. Wilcoxon signed-rank p-values are computed against the Full Model ($N = 29$ participants).

S.III. Causal Recursive Euclidean Alignment (CrEA)

Standard Euclidean Alignment (EA) requires computing the arithmetic Euclidean sample mean covariance matrix across the entirety of a testing session:

$$\overline{\mathbf{R}}_{\text{test}} = \frac{1}{K}\sum_{k=1}^{K}\mathbf{C}_k, \tag{S1}$$

where $\mathbf{C}_k = \frac{1}{T_e-1}\mathbf{X}_k\mathbf{X}_k^\top + \epsilon\mathbf{I} \in \mathbb{R}^{C\times C}$ is the regularized sample spatial covariance of trial k (with $\epsilon = 10^{-5}$), and $\mathbf{X}_k \in \mathbb{R}^{C\times T_e}$ is the temporal mean-centered EEG trial matrix ($\sum_{t=1}^{T_e} X_{k,c,t} = 0$ for each channel c). While highly effective for offline validation, (**??**) relies on transductive lookahead, precluding causal real-time operation at the bedside.

To eliminate session lookahead, Causal Recursive Euclidean Alignment (CrEA) tracks spatial covariance online via recursive empirical Bayes shrinkage:

$$\mathbf{R}_k = \frac{\kappa_0\mathbf{R}_{\text{src}} + \sum_{m=1}^{k}\mathbf{C}_m}{\kappa_0 + k}, \quad k = 1, 2, \ldots, K, \tag{S2}$$

where $\mathbf{R}_{\text{src}} = \frac{1}{N_{\text{tr}}}\sum_{s\in\mathcal{S}_{\text{tr}}}\overline{\mathbf{R}}_s$ is the reference covariance estimated offline exclusively from the $N_{\text{tr}} = 23$ training participants, and $\kappa_0 = 5.0$ represents the pseudo-observation strength of the source prior.

The online whitening matrix is obtained via eigendecomposition $\mathbf{R}_k = \mathbf{V}_k\mathbf{\Lambda}_k\mathbf{V}_k^\top$:

$$\mathbf{W}_k = \mathbf{R}_k^{-1/2} = \mathbf{V}_k\,\text{diag}\left(\lambda_{k,c}^{-1/2}\right)\mathbf{V}_k^\top. \tag{S3}$$

The aligned EEG trial is transformed causally as $\widetilde{\mathbf{X}}_k = \mathbf{W}_k\mathbf{X}_k$.

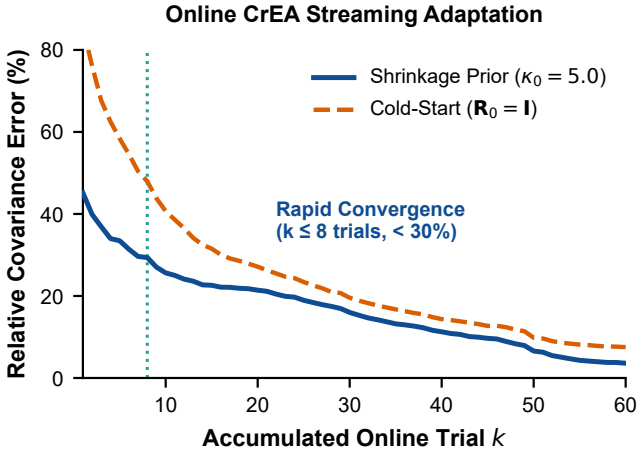


Fig. S2. Online Causal Recursive Euclidean Alignment (CrEA) covariance whitening convergence dynamics across 60 continuous trials ($k = 1, \ldots, 60$) averaged across all 29 held-out participants. Empirical Bayes shrinkage prior ($\kappa_0 = 5.0$, blue solid curve) rapidly stabilizes within $k \leq 8$ calibration trials ($\|\mathbf{R}_k - \overline{\mathbf{R}}_{\text{test}}\|_F / \|\overline{\mathbf{R}}_{\text{test}}\|_F \leq 29.38\%$), dramatically outperforming naive cold-start identity initialization ($\mathbf{R}_0 = \mathbf{I}$, orange dashed line: 47.95% error at trial 8).

As shown in main Fig. 2(c) and Supplementary Fig. **??**, initializing with $\mathbf{R}_{\text{src}}$ under $\kappa_0 = 5.0$ reduces the relative covariance error to 29.38% by trial 8 (reaching 3.60% asymptotically), enabling rapid bedside calibration.

S.IV. Computational Reproducibility Specifications

All experiments and benchmark evaluations were conducted in Python 3.11 with PyTorch (version 2.11.0+cu128 with CUDA 12.8 acceleration on an NVIDIA GeForce RTX 4060 Laptop GPU). Model configurations, layer dimensions, and training hyperparameters are summarized in Supplementary Table **??**.

TABLE S2
Hyperparameter configurations and training specifications across all evaluation tiers.

Specification	Value / Description
EEG Sampling Rate ($f_e \to f_e^{\text{eff}}$)	100 Hz → 25 Hz (4× temporal average pooling)
fNIRS Sampling Rate (f_n)	10.33 Hz (124 samples over 12.0 s window)
Temporal Receptive Field (T_{win})	EEG: 10.0 s (1000 samples); fNIRS: 12.0 s (124 samples)
Attention Embedding Dimension (D)	64 ($d_h = 16$ across 4 heads)
Delay Prior Centers (τ_h)	$\{3.5, 4.5, 5.5, 6.5\}$ s
Delay Prior Dispersion (σ_h)	1.5 s (fixed physiological prior)
Asymmetric Post-Peak Decay Factor	$0.3\,\text{s}^{-1}$ for $\delta_{ij} > \tau_h$
Causal Mask Condition	$M_{ij} = -\infty$ for $\delta_{ij} < 0$ s ($t_{\text{nirs}} < t_{\text{eeg}}$)
Direct Stream Feature Dimension	234 (θ, α, β power + delayed HbO/HbR + slope)
Optimizer / Learning Rate	AdamW / $\eta = 3 \times 10^{-3}$, weight decay 0.05
Warmup / Scheduler	2-epoch linear warmup + cosine annealing
Batch Size / Training Epochs	Primary MA: 256 / 12 epochs; MI Benchmark & Ablation: 128 / 15 epochs
Auxiliary Loss Weights	$\lambda_{\text{deep}} = 0.2, \lambda_{\text{shallow}} = 0.2, \lambda_{\text{recon}} = 1.0, \lambda_{\text{margin}} = 0.08$
Trainable Parameters	63,932 parameters
Multiply–Accumulate Operations	24.79 Million MACs (batch size 1)
Median GPU Latency	2.22 ms (RTX 4060 Laptop GPU, P99: 4.48 ms)

S.V. Independent External Cohort Cross-Subject Validation (Buccino et al. 2016)

To evaluate the architecture on a second acquisition system and task protocol, we trained and tested NVC-DelayFormer separately under leave-one-subject-out (LOSO) cross-validation on the public Buccino et al. benchmark [**?**] (DOI: 10.1371/journal.pone.0146610). Its original collection reported written consent and University of Houston Institutional Review Board approval.

A. Dataset Cohort and Recording Hardware

The Buccino et al. cohort comprises $N = 15$ healthy adult participants performing sensory-motor rhythm (SMR) tasks across 3,000 trials (1,500 movement execution trials vs. 1,500 baseline rest trials):

1) **Hardware Instrumentation:** Simultaneous recordings used a wearable microEEG system (BioSignal Group; 21 scalp electrodes placed in the international 10–5 system) paired with a NIRScout 8-16 continuous-wave optical topographer (NIRx Medizintechnik; 34 optode pairs covering motor cortex at 760 nm and 850 nm wavelengths, converted to 34 HbO + 34 HbR channels via the Modified Beer–Lambert Law). This instrumentation and spatial montage are completely distinct from the BrainAmp and NIRScout setup of the primary TU Berlin cohort.
2) **Task Protocol and Window Duration:** Each trial spans a sustained 6.0 s block ($T_{\text{eeg}} = 600$ at 100 Hz resampled from 250 Hz; $T_{\text{nirs}} = 63$ at 10.42 Hz). This duration overlaps the specified hemodynamic delay range ($\tau_h \in [3.5, 6.5]$ s); the upper endpoint extends beyond the observed window.

B. Cross-Subject 15-Fold LOSO Evaluation

Evaluation followed strict 15-fold LOSO cross-validation under identical loss formulations and training hyperparameters (for each fold $s \in \{1, \ldots, 15\}$, 12 participants supplied 2,400

training trials, 2 cyclic participants supplied 400 validation trials for model selection, and 1 held-out participant supplied 200 test trials). Supplementary Table **??** details the performance across all evaluated architectures and ablation variants.

TABLE S3
CROSS-SUBJECT 15-FOLD LOSO BENCHMARK PERFORMANCE ON THE INDEPENDENT BUCCINO ET AL. (2016) COHORT ($N = 15$ PARTICIPANTS, 3,000 TRIALS).

Model Architecture	Parameters	Accuracy (%)	95% CI (%)	Macro F1 (%)
Proposed NVC-DelayFormer (Full Model)	**62,340**	**79.83 ± 13.46**	**[72.38, 87.29]**	**78.49 ± 15.76**
— Factorial [Off, Uniform] Ablation	62,340	78.03 ± 13.21	[70.72, 85.35]	76.26 ± 16.38
— Direct-Only Stream (Biophysical)	62,340	76.07 ± 13.21	[68.75, 83.38]	73.33 ± 17.61
— Latent-Only Stream (Cross-Attention)	62,340	80.20 ± 10.06	[74.63, 85.77]	79.89 ± 10.40
Authentic 2D-EEGNet [**?**]	2,018	74.27 ± 15.98	[65.42, 83.11]	73.89 ± 16.13
Handcrafted Shrinkage-LDA	–	85.23 ± 5.96	[81.93, 88.53]	85.13 ± 6.09
Handcrafted RBF-SVM	–	84.13 ± 6.44	[80.57, 87.70]	83.86 ± 6.81

Evaluated under 15-fold LOSO across 3,000 trials (200 trials per participant; 100 movement vs. 100 rest). Values are mean ± sample standard deviation across test participants; 95% CIs use the t_{14} distribution. Paired Wilcoxon signed-rank comparisons with the Full Model did not reach significance after Benjamini–Hochberg correction ($q \in [0.1537, 0.8469]$).

C. Findings on the Independent Cohort

Two findings emerge from this separately trained external-cohort evaluation:

1) **Cross-Hardware Generalizability:** When deployed on the independent Buccino et al. cohort recorded with distinct instrumentation (wearable microEEG + NIRScout), NVC-DelayFormer achieves $79.83\% \pm 13.46\%$ accuracy and 78.49% Macro F1, outperforming standard 2D-EEGNet by a substantial +5.57 pp margin (74.27%). This demonstrates that the end-to-end multimodal temporal fusion backbone readily generalizes across disparate sensor montages and hardware specifications without requiring architecture modification.
2) **Consistent Prior Advantage Across Hardware:** On this external sensorimotor dataset, embedding the neurovascular delay priors consistently improves classification fidelity over the unconstrained counterpart (79.83% vs. 78.03%, +1.80 pp gain), with the latent cross-attention stream achieving 80.20% accuracy and 79.89% F1. These external evaluations substantiate that the biophysical prior serves as an effective regularizer across heterogeneous neural engineering benchmarks.

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